# AstroQuark Engineering Report

## 1 Project Overview

### 1.1 Motivation and Objectives

I started this project to study how a planetary rover can move over uneven ground and record what happens during a traverse. The work began with a mechanical design in SolidWorks and progressed to a simulated rover that follows a lunar-style course, records its motion and stops at a predefined observation point.

### 1.2 Scope of the Demonstration

The demonstration focuses on low-speed mobility, chassis pitch and the relationship between wheel rotation and distance traveled. The original project brief also identified regolith characterization, thermal variation, radiation, energy and obstacle sensing as longer-term goals. This version provides the mechanical and simulation foundation for those investigations.

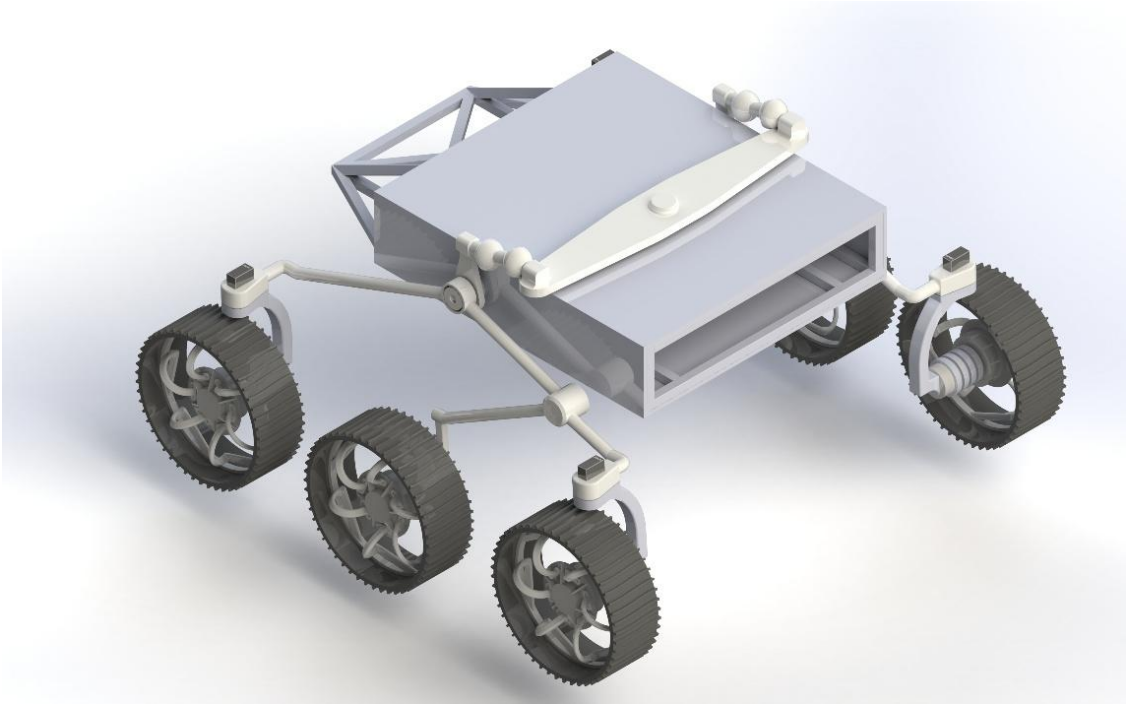


**Figure 1. Overall mechanical design of the six-wheel rover.**

# 2 Mechanical Design

## 2.1 Research and Design Selection

After conducting detailed research into planetary rover designs, I selected a six-wheel rocker–bogie suspension. NASA's Curiosity Mars rover uses a similar six-wheel rocker-bogie layout. I chose it because the articulated links allow the wheels to follow changes in ground height, helping maintain contact on uneven terrain. This made it suitable for studying low-speed mobility and chassis pitch on a rocky course.

## 2.2 SolidWorks Parts and Assembly

I worked with the mechanical assembly in SolidWorks, which includes the chassis and plates, rocker members, wheels, wheel connectors and steering-related components. The CAD also includes a differential component, ball rods and sockets. On each side, the rocker pivots relative to the chassis and carries the front wheel and the bogie pivot. The bogie carries the middle and rear wheels, letting them move relative to the rocker as terrain height changes.

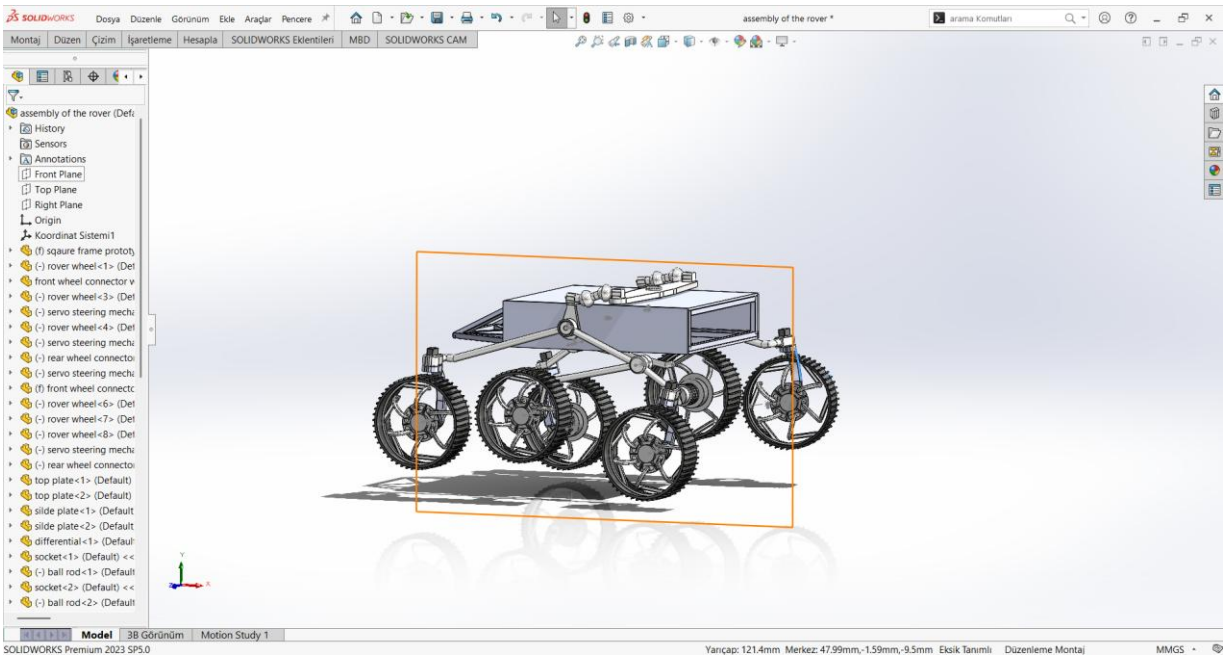



**Figure 2. SolidWorks assembly showing the component tree, chassis and wheel linkages.**

The suspension helps maintain wheel contact over changing terrain. Contact and chassis stability still depend on the slope, wheel placement and loading.

# 3 From CAD to Simulation

## 3.1 Links Joints and Geometry

I exported the SolidWorks assembly to URDF and prepared an SDF model for Gazebo. The model separates the body, suspension and wheels into 11 links. Four suspension pivots and six wheel joints connect those links. The simulated wheels use cylindrical collision geometry with a radius of approximately 0.131936 m, which is also the radius used in the motion calculation.

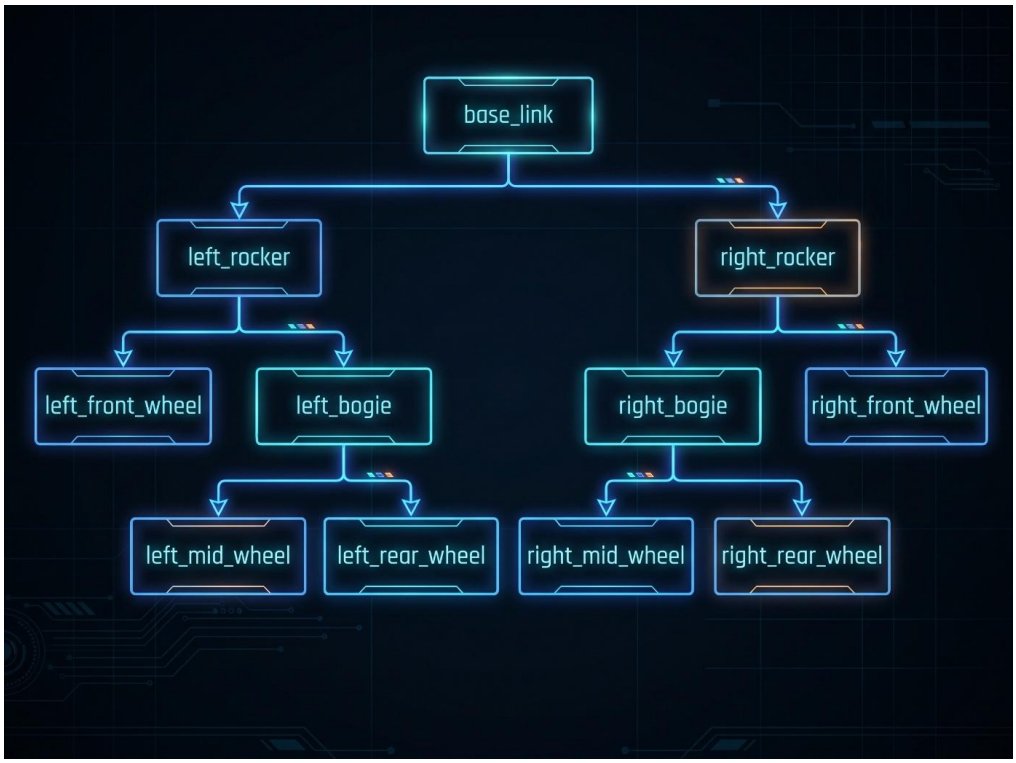



**Figure 3. Link hierarchy from the chassis to the rockers, bogies and wheels.**

## 3.2 Flat Ground and Drive Control

The first simulated motion was a straight run on flat ground. A wheel velocity controller applied a shared angular-speed command to the six wheels. This provided a baseline for checking model motion and the response to a stop command. The simulated drivetrain used in the recorded runs represents the articulated suspension with a common drive command. Independent steering and mechanical coupling through the CAD differential are outside these runs.

# 4 Lunar Environment and Data Collection

## 4.1 Lunar Course in Gazebo

I prepared a separate lunar world with a downward gravitational acceleration of 1.62 m/s². The course starts on level ground, rises up an 8° ramp and finishes on a plateau approximately 0.348 m above the starting surface. Rocks with collision geometry sit beside the route. A visual crater, surface scatter and distant ridges provide scene context. The course uses simplified solid contact for repeatable mobility tests.

## 4.2 ROS 2 Integration

ROS 2 Jazzy exchanges commands and measurements with Gazebo Harmonic through ros_gz_bridge. The control script sends a wheel-speed command. The recorder uses joint states, an IMU mounted on the body, Gazebo reference pose and simulation time to log a complete traverse. The IMU orientation is converted from a quaternion to chassis pitch in degrees, with positive pitch indicating that the nose is up.

| Signal | Role in the experiment |
|---|---|
| Wheel command | Sends a common target angular speed to the six wheel joints. |
| Joint states | Provides measured wheel rotation for the kinematic estimate. |
| Body IMU | Provides orientation for calculating chassis pitch. |
| Reference pose and clock | Provide body position and time for measurement and logging. |

## 4.3 Automated Traverse

The rover starts on a predefined straight route. It reduces its wheel-speed command near the ramp and again on the approach to the plateau observation point. The controller sends a zero command when the reference position reaches the target at x = 5.25 m. Recording continues after that command so the final stop can be checked in the measured data.

# 5 Motion Model and Results

## 5.1 Wheel Based Motion Estimate

I estimated forward speed using $v = r \times \bar{\omega}$, where r is the simulated wheel radius and $\bar{\omega}$ is the mean magnitude of the six measured wheel rates. I integrated that speed over simulation time using the trapezoidal rule. For comparison, the reference distance sums the three-dimensional position increments of the rover body in Gazebo. This comparison shows how closely the rolling approximation matches the recorded motion on this course.

## 5.2 Ramp Traversal Results

| Command (rad/s) | Duration (s) | Peak pitch (°) | Gazebo (m) | Estimate (m) | Difference | Check |
|---|---|---|---|---|---|---|
| 1.5 | 34.40 | 7.88 | 5.286 | 5.309 | 0.43% | Pass |
| 1.8 | 29.00 | 7.89 | 5.289 | 5.315 | 0.49% | Pass |
| 2.2 | 23.90 | 7.88 | 5.286 | 5.318 | 0.60% | Pass |

All three runs crossed the ramp, reached the observation point and came to rest. The highest recorded uphill pitch was about 7.89°. The largest final distance difference was 0.60% across the three runs. These results describe the simulated course and its contact settings. The archived CSV logs, verification results and plots retain the underlying measurements.

## 5.3 Engineering Takeaways

The project connected a CAD assembly to a working simulation with a complete command, measurement and analysis workflow. The recorded pitch shows when the body responds to the ramp, and the distance comparison provides a baseline for investigating slip and more complex terrain in later runs.

## 5.4 Speed Pitch and Distance

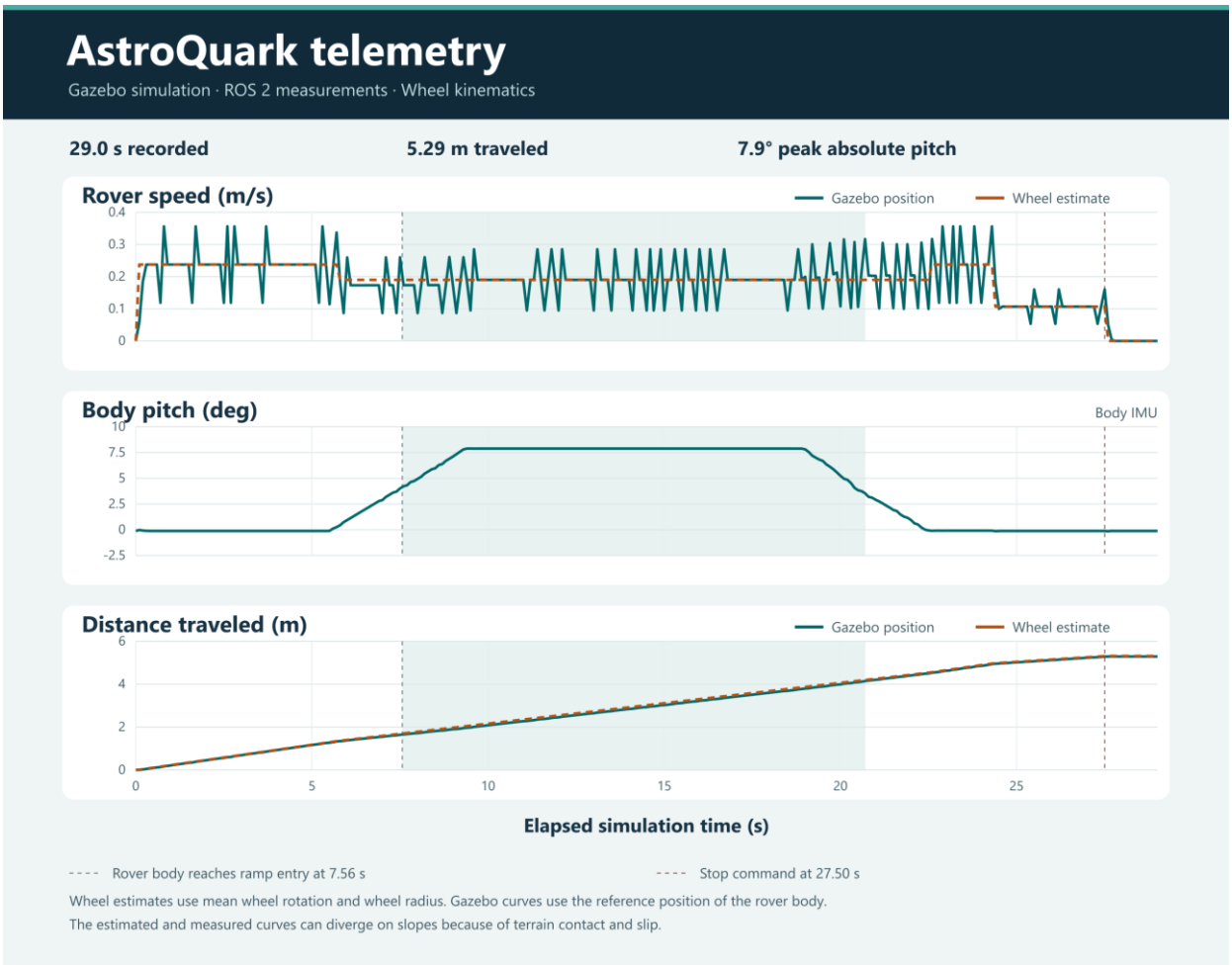



**Figure 4. Speed, body pitch and distance for the 1.8 rad/s run.**

The speed and distance plots compare values estimated from wheel rotation with measurements derived from the rover position in Gazebo. The middle plot shows body pitch from the IMU.